

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA17

COURSE OUTCOME COVERED

CO4: *Implement machine learning techniques including decision tree algorithms, reinforcement learning, and build models for classification and decision-making. (BL3)*

Assessment Tool Weightage: 50%

Duration : 1 hour

QUESTIONS: 4

Total Marks:40

Annexure A

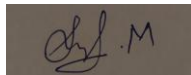
ARTICLE REVIEW

Code of Conduct:

I SOUNDARIYA M(192524342) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



ARTICLE REVIEW

INSTRUCTIONS

Select any ONE peer-reviewed research article (published after 2019) from IEEE Xplore, Springer, Elsevier, or ACM Digital Library related to any of the following topics: *Inductive Learning, Decision Trees, Statistical Learning Methods (Naïve Bayes / Logistic Regression / SVM), or Reinforcement Learning applications*.

Submit: (i) Article citation details (ii) PDF or valid DOI link (iii) Written review as per the tasks below.

Task 1: Article Summary

(a) Provide the full citation of the article (Author(s), Title, Journal/Conference, Year, DOI). State the domain/application area and the specific ML problem addressed.

(b) Summarise the key objective of the article in your own words (150–200 words). Clearly state the research gap the authors identified and how they proposed to fill it.

Task 2: Methodology Analysis

(a) Explain the ML technique(s) used in the article (e.g., Decision Tree variant, RL algorithm, Statistical model). Describe the dataset used—size, features, source, and how it was prepared.

(b) Map the technique to CO 4 topics (Inductive Learning / Decision Trees / Statistical Learning / RL). Identify one methodological strength and one limitation in the authors' approach.

Task 3: Results & Critical Evaluation

(a) Report the key results (accuracy, F1-Score, AUC-ROC, reward convergence, etc.) as presented by the authors. Create a comparative table if multiple models are benchmarked.

(b) Critically evaluate the results: Are the reported metrics realistic? Are there potential biases (dataset, evaluation protocol)? What additional experiments would strengthen the findings?

(c) Discuss real-world applicability: Where can this technique be deployed? What are the challenges in scaling the solution to industry (data availability, compute, ethical issues)?

Task 4: Reflection & Learning Outcome

(a) State three key insights you gained from reading the article that deepen your understanding of CO 4 topics. Connect each insight to a specific concept from the course syllabus.

(b) If you were to extend this research, propose one novel experiment or enhancement (different dataset / improved algorithm / new application domain). Justify its feasibility and expected impact.

RUBRICS FOR CALCULATION:

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Article	20%	Demonstrates clear and in-depth understanding of the article's purpose, concepts, and arguments	Shows good understanding with minor gaps	Partial understanding; misses key ideas	Lacks understanding of the article
Summary	20%	Provides concise and accurate summary highlighting all key points	Covers most key points with minor omissions	Incomplete or partially accurate summary	Poor or incorrect summary
Critical Analysis	25%	Provides insightful critique with strong reasoning and evaluation	Provides reasonable analysis with some justification	Superficial analysis; limited evaluation	No meaningful analysis
Use of Evidence	15%	Effectively uses examples, data, or references to support review	Uses some supporting evidence with minor gaps	Limited or weak supporting evidence	No supporting evidence provided
Organization	20%	Well-structured, clear, and coherent writing with proper language and flow	Generally clear with minor issues	Poor organization; lacks clarity	Disorganized and difficult to understand

TASK 1: ARTICLE SUMMARY

(a) Full Citation, Domain and ML Problem

Article Title:

Predictive model and feature importance for early detection of type II diabetes mellitus

Authors:

Eric Adua, Emmanuel Awuni Kolog, Ebenezer Afrifa-Yamoah, Bright Amankwah, Christian Obirikorang, Enoch Odame Anto, Emmanuel Acheampong, Wei Wang, and Antonia Yarney Tetteh

Journal:

Translational Medicine Communications

Year:

2021

Volume:

6, Article 17

DOI:

<https://doi.org/10.1186/s41231-021-00096-z>

Domain/Application Area:

Healthcare, Medical Diagnosis and Diabetes Prediction

ML Problem:

The article addresses the problem of early prediction of Type II Diabetes Mellitus (T2DM). It is a binary classification problem in which patients are classified as diabetic or non-diabetic using clinical and biochemical features.

The researchers compared four machine learning algorithms:

1. Naïve Bayes
2. K-Nearest Neighbour (KNN)
3. Support Vector Machine (SVM)
4. Decision Tree

(b) Key Objective and Research Gap

The main objective of the article was to develop machine learning models for the early detection of Type II Diabetes Mellitus and identify the clinical features that are most important for prediction. The researchers focused on a Ghanaian population because diabetes risk can be influenced by demographic, lifestyle and biological factors, and models developed for one population may not perform equally well in another population.

The research gap was that conventional statistical approaches may not adequately capture complex relationships between different clinical variables. The authors therefore investigated multiple machine learning algorithms and compared their ability to predict T2DM. Another important gap was the need to identify the most influential predictors rather than only developing a prediction model.

The authors used clinical and biochemical data and compared Naïve Bayes, KNN, SVM and Decision Tree algorithms. Recursive Feature Elimination was also used to determine feature importance. The study aimed to support early diabetes detection and improve diabetes risk screening, especially in healthcare environments where medical resources may be limited.

TASK 2: METHODOLOGY ANALYSIS

(a) ML Techniques and Dataset

The article used four supervised machine learning algorithms:

1. Naïve Bayes:

Naïve Bayes is a probabilistic statistical learning algorithm based on Bayes' theorem. The study used Gaussian Naïve Bayes for classification.

2. Decision Tree:

Decision Tree is an inductive learning technique that divides patient records into branches based on feature values and produces a final classification at the leaf nodes.

3. Support Vector Machine (SVM):

SVM finds an optimal decision boundary that separates diabetic and non-diabetic patients. The study used an RBF kernel.

4. K-Nearest Neighbour (KNN):

KNN classifies a new patient based on the classes of nearby patients in the feature space.

Dataset:

The study used 438 participants:

- 219 Type II Diabetes Mellitus patients
- 219 healthy control participants

The T2DM patients were recruited from Komfo Anokye Teaching Hospital in Ghana, while healthy controls were recruited from communities in the Kumasi metropolitan area.

The dataset contained 11 attributes:

1. Age
2. Body Mass Index (BMI)
3. Systolic Blood Pressure (SBP)
4. Diastolic Blood Pressure (DBP)
5. HbA1c
6. Fasting Blood Sugar (FBS)
7. Total Cholesterol (TC)
8. Triglycerides (TG)
9. HDL Cholesterol
10. LDL Cholesterol
11. T2DM Class

The first 10 variables were used as predictors and T2DM status was the target variable.

Data Preparation:

- The dataset was checked for errors and outliers.
- 17 missing observations were identified.
- Missing values were estimated using the Expectation-Maximization algorithm.
- Predictive variables were scaled between 0 and 1.
- Recursive Feature Elimination (RFE) was used for feature ranking.
- The dataset was divided into 80% training and 20% testing data.
- Equal numbers of diabetic and control cases were maintained.

(b) Mapping to CO 4 Topics

Inductive Learning:

The algorithms learned patterns from labelled patient records and used them to predict the diabetes status of unseen patients.

Decision Trees:

Decision Tree was one of the machine learning algorithms used for predicting T2DM.

Statistical Learning:

Naïve Bayes is a statistical learning method based on probability and Bayes' theorem.

Reinforcement Learning:

Reinforcement Learning was not directly used in this article.

Methodological Strength:

A major strength is that the researchers compared multiple machine learning algorithms using the same dataset. This allows their predictive performance to be compared fairly. The use of Recursive Feature Elimination also helped identify the most important clinical predictors.

Methodological Limitation:

The dataset contained only 438 participants and was collected from a specific Ghanaian population. Therefore, the results may not generalize to other populations. The study also used a single train-test split instead of repeated cross-validation.

TASK 3: RESULTS & CRITICAL EVALUATION

(a) Key Results

The researchers compared four machine learning models.

Model	Accuracy	AUC-ROC
Naïve Bayes	87%	0.87
SVM	84%	0.84
KNN	83%	0.85
Decision Tree	81%	0.81

Naïve Bayes achieved the best overall accuracy and AUC-ROC.

For diabetic patients, Naïve Bayes detected 82% of the test cases and 93% of the detected cases were correctly predicted. The F1-score for diabetic patients was 87%.

The feature importance analysis identified the following top three predictors:

1. HbA1c
2. Total Cholesterol (TC)
3. BMI

The least important features included Age, HDL Cholesterol and LDL Cholesterol according to the feature-ranking analysis.

(b) Critical Evaluation

The reported results are reasonably realistic because the models achieved accuracies between 81% and 87%, rather than extremely high values close to 100%.

However, there are some limitations:

1. Small Dataset:

The study used only 438 participants. A larger dataset would provide more reliable results.

2. Population Bias:

The participants were from Ghana. Differences in lifestyle, genetics, healthcare systems and population characteristics may affect the performance of the model in other countries.

3. Sampling Bias:

The study used purposive sampling for T2DM patients and convenience sampling for healthy controls. This may not fully represent the general population.

4. Evaluation Protocol:

The authors used an 80:20 train-test split. Repeated k-fold cross-validation could provide a more reliable estimate of model performance.

Additional experiments that could strengthen the research include:

- Using a larger multi-centre dataset.
- Performing 5-fold or 10-fold cross-validation.
- Testing the model on an independent external dataset.
- Reporting confidence intervals.
- Testing ensemble models such as Random Forest and Gradient Boosting.

- Performing fairness analysis across demographic groups.
- Evaluating sensitivity and specificity in greater detail.

(c) Real-World Applicability

The proposed approach can be used in:

- Hospitals
- Primary healthcare centres
- Diabetes screening programmes
- Electronic Health Record systems
- Rural healthcare centres
- Telemedicine systems
- Public-health diabetes surveillance

The model could help healthcare professionals identify patients who have a high probability of diabetes and may require further clinical testing.

Challenges in Industry Deployment:

1. Data Availability:

Large, high-quality and representative medical datasets are required.

2. Data Quality:

Missing values, measurement errors and inconsistent medical records can affect predictions.

3. Generalization:

A model trained using Ghanaian patients may not perform equally well on patients from other populations.

4. Privacy:

Patient medical information must be securely stored and protected.

5. Explainability:

Doctors need understandable reasons for model predictions.

6. Clinical Validation:

The model must be tested in real clinical environments before deployment.

7. Ethical Issues:

Incorrect predictions can result in missed diagnosis or unnecessary medical testing.

TASK 4: REFLECTION & LEARNING OUTCOME

(a) Three Key Insights

Insight 1: Inductive Learning

The article helped me understand that inductive learning learns general patterns from labelled examples. In this study, patient records containing clinical features were used to learn patterns associated with Type II Diabetes Mellitus.

Insight 2: Comparison of Classification Models

I learned that different machine learning algorithms can produce different results on the same dataset. In this study, Naïve Bayes achieved 87% accuracy, while Decision Tree achieved 81% accuracy.

Insight 3: Feature Importance

The article demonstrated the importance of identifying the variables that contribute most to prediction. HbA1c, Total Cholesterol and BMI were identified as the top three important features.

(b) Proposed Research Extension

I would extend the research by testing the models on a larger multi-country diabetes dataset and comparing the existing algorithms with an ensemble method such as Random Forest or Gradient Boosting.

The experiment would use:

- The original Ghanaian dataset.
- An external diabetes dataset from another population.
- 5-fold or 10-fold cross-validation.
- External validation.
- Accuracy, Precision, Recall, F1-Score and AUC-ROC.
- Fairness analysis across demographic groups.

This experiment is feasible because the original research already provides a complete machine learning workflow involving preprocessing, feature selection, model training and evaluation.

Using a larger and more diverse dataset would help determine whether the model can generalize to different populations. It could improve the reliability of diabetes prediction and support the development of a more robust clinical decision-support system.